Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Mountain View, CA

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) building LLM-agent evaluation pipelines, backend automation, and cloud infrastructure with Java, Python, AWS, and Kubernetes. Current research designs closed-loop evaluation for autonomous agent systems, benchmarking LLM-driven pipelines against published baselines.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.

Agent & LLM Systems: Gemini, Vertex AI, MCP, LangChain, agent orchestration, tool-calling, RAG, eval pipelines.

Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs.

Databases: PostgreSQL, MySQL.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, SonarQube, Azure, Google Cloud Platform.

AI & Data: PyTorch, TensorFlow, scikit-learn, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed closed-loop evaluation for a two-stage memory-augmented LLM agent pipeline automating Android GUI bug reproduction, benchmarked against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Built device-automation agent harness (uiautomator2/ADB) with stall detection and history-windowed decision loop; measured **25.7x latency reduction** (39.4s vs 1012.1s avg) while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services with full auditability.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** with reliable rollback.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**; developed **Go + Terratest** framework catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction and cross-payer comparison, consumable by external AI agents; validated every tool boundary with Zod schemas.

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).